

BLUEPRINTIQ TECHNICAL MEMORANDUM

UNCERTAINTY-AWARE BLUEPRINT-TO-BOQ INTELLIGENCE | DETAILED ANALYSIS REPORT

STATUTORY ENGINEERING NOTICE: BlueprintIQ provides automated preliminary computer-vision geometry extraction and material estimations. This document does not replace a licensed professional engineer, architect, quantity surveyor, or municipal authority approval. All quantities must be field-verified before procurement.

1. PROJECT & DRAWING METADATA

Project Name:	Phase 2 Verification Villa	Building Type:	Residential
Floors:	3	Unit System:	METRIC
Target Area:	220.0 m²	Soil Type:	Not Provided
Drawing Scale:	Unknown	Concrete Grade:	M20
Wall Thickness:	230 mm	Overall Confidence:	MEDIUM
Overall Confidence Score:	87% (Estimated AI/CV Analysis)	Estimation Basis:	Computer-vision floor-plan extraction & quantity modeling

2. CONSOLIDATED MATERIAL ESTIMATES (UNCERTAINTY-AWARE)

MATERIAL	ESTIMATED	RANGE	CONF.	CALCULATION BASIS
Cement (OPC/PPC 50kg Bags)	3,416.0 bags	3,143 - 3,757	MEDIUM	Concrete (1021.9 bags) + Masonry (1531.3 bags) + Plaster (862.6 bags)
River Sand / M-Sand	515.3 m³	464 - 577	MEDIUM	Concrete (53.2m³) + Masonry (319.0m³) + Plaster (143.0m³)
Coarse Aggregate (20mm & 10mm Graded)	133.2 m³	120 - 146	MEDIUM	RCC Concrete (106.4m³) + PCC Bed (26.8m³)
Modular Clay / Fly Ash Bricks	445,210.0 nos	409,593 - 489,731	HIGH	848.02 m³ net masonry x 500 bricks/m³ + 5% breakage allowance
Reinforcement Steel (Fe500 TMT Rebars)	9,481.0 kg	8,059 - 11,852	MEDIUM	Slabs (6217.0 kg at 0.8% vol) + Beams/Columns (3264.0 kg at 1.5% vol)
Vitrified Floor Tiles (600x600mm)	34.7 m²	33 - 38	HIGH	Net floor area 32.1m² + 8% cutting and breakage allowance
Emulsion Paint & Primer (2 Coats)	859.6 litres	774 - 988	MEDIUM	Total plaster & ceiling area 8596.0m² / coverage 10 m²/Litre

3. DETAILED BILL OF QUANTITIES (CIVIL & FINISHES)

CODE	DESCRIPTION	QTY	UNIT	RANGE	CONF.
CIV-01	Earthwork excavation in foundation trenches	434.52	m³	304.2 - 630.0	LOW
CIV-02	Plain Cement Concrete (PCC 1:4:8) in foundation bed (100mm)	31.49	m³	28.3 - 36.2	MEDIUM
CIV-03	Reinforced Cement Concrete (M20) for Slabs, Beams, Columns	126.72	m³	116.6 - 141.9	MEDIUM
CIV-04	Brick Masonry in Cement Mortar (1:6) with Modular Bricks	848.02	m³	763.2 - 932.8	HIGH
CIV-05	Cement Plastering (12mm Internal 1:6 + 18mm External 1:4)	7935.98	m²	7380.5 - 8729.6	MEDIUM
CIV-06	Flooring with Vitrified Tiles including 8% wastage	34.67	m²	32.9 - 37.4	HIGH

4. BLUEPRINT ANOMALIES & CODE COMPLIANCE FINDINGS

ID	SEV.	FINDING & EVIDENCE	RECOMMENDED ACTION
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ISSUE-001	HIGH	Drawing Scale Not Confirmed or Uncalibrated Drawing scale specified as 'Unknown' without dimension calibration points. <i>Standard: IS 1200 / ISO 128 Architectural Drawing Presentation Rules</i>	Perform scale calibration against a known reference dimension on the drawing before procurement. Status: Verification Required
ISSUE-002	MEDIUM	Potential Habitable Area Constraint: LIVING & DINING Measured area: 3.8 m² vs Recommended threshold: 9.5 m² <i>Standard: NBC 2016 Part 3 Clause 4.2.1 (Minimum Habitable Room Area)</i>	Review room layout with a licensed architect to confirm local municipal development control guidelines. Status: Verification Required
ISSUE-003	MEDIUM	Staircase Riser and Tread Detail Missing on 2D Plan No textual riser/tread callouts (e.g. '18 Risers @ 165mm') identified in staircase zone. <i>Standard: IBC Section 1011.5.2 & NBC 4.8.1</i>	Request sectional drawing / stair detailing schedule for precise concrete and reinforcement calculation. Status: Verification Required

5. LOCAL AI (GEMMA 2B) SYNTHESIS & REASONING

Executive Synthesis: Preliminary architectural evaluation performed for Residential project. Detected 3 potential dimensional or compliance considerations (1 require urgent review). Soil parameter was 'Not Provided', requiring geotechnical confirmation before finalizing foundation estimates.

Uncertainty Assessment: Overall analysis confidence is rated as MEDIUM. Confidence is constrained by 2D plan annotations and missing structural section schedules.

6. MISSING PARAMETERS & ENGINEERING ASSUMPTIONS

- Missing Input Parameters:**
- **Soil Type:** Not Provided — *Foundation trench excavation, footing depth, and sub-base quantities cannot be verified without soil bearing capacity (SBC).*
 - **Scale Calibration:** Uncalibrated — *Drawing scale relies on nominal text ratio ('1:100') rather than two-point physical dimension calibration. Coordinate errors may magnify calculated areas by 5-10%.*
- Applied Engineering Assumptions:**
- Excavation depth assumed at nominal 1.2m for medium dense soil; footing sizes subject to soil report.
 - Graphic scale ratio Unknown assumed uniformly proportional across X and Y axes.
 - Reinforcement steel estimated via standard IS 456 thumb rules (0.8% for slabs, 1.5% for framed elements).